

# Hridoy A. Hazarika

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## EDUCATION

### Tezpur University

*Integrated Msc - Mathematics*

Tezpur, Assam, India

*Aug. 2024 – May 2029*

## EXPERIENCE

### Undergraduate Research Student

*Tezpur University*

Tezpur, Assam, India

*Aug. 2024 – Present*

- Studied advanced topics in linear and abstract algebra, real analysis, number theory and combinatorics beyond coursework.
- Prepared mathematical documents, assignments, and reports using  $\text{\LaTeX}$ .
- Used MATLAB for computational exploration and visualization of mathematical concepts.

### Research Intern

*Madhabdev Univeristy*

Lakhimpur, Assam, India

*Jul. 2026 – Aug. 2026*

- Conducted a research internship under the supervision of Dr. Abinash Sarma, Department of Mathematics, Madhabdev University.
- Explored  $q$ -Series, Partition Theory and Theta Functions, with emphasis on their fundamental concepts, identities, and theorems.
- Developed a deeper understanding of the connections between  $q$ -Series, partitions, and classical results in number theory.

### Research Intern

*Indian Institute of Technology Guwahati*

Guwahati, Assam, India

*Jun. 2026 – Jul. 2026*

- Completed a summer research internship (SIP2026 Programme) under the supervision of Dr. Arup Chattopadhyay.
- Studied the geometric foundations of inner product spaces, including orthogonal projections, linear functionals, hyperplanes, and Hilbert spaces.
- Prepared a comprehensive research report in  $\text{\LaTeX}$  with rigorous mathematical exposition and proofs.

### Independent Research Study

*Analytic Number Theory*

Remote

*May 2026 – Present*

- Independently studying analytic number theory with emphasis on  $q$ -series, partition theory, and modular forms.
- Reading research papers and classical texts to build foundations for future research.

## TECHNICAL SKILLS

**Programming:** C, Python, C++(Basic), SQL

**Web Technologies:** HTML, CSS, JavaScript

**Typesetting:**  $\text{\LaTeX}$  (Overleaf, mathematical typesetting, TikZ)

**Mathematical Software:** MATLAB, Maple, GeoGebra

**Productivity:** Microsoft Office Tools, Adobe Tools, Canva

**Version Control:** Git, GitHub

**Operating Systems:** Windows, Linux (Basic)